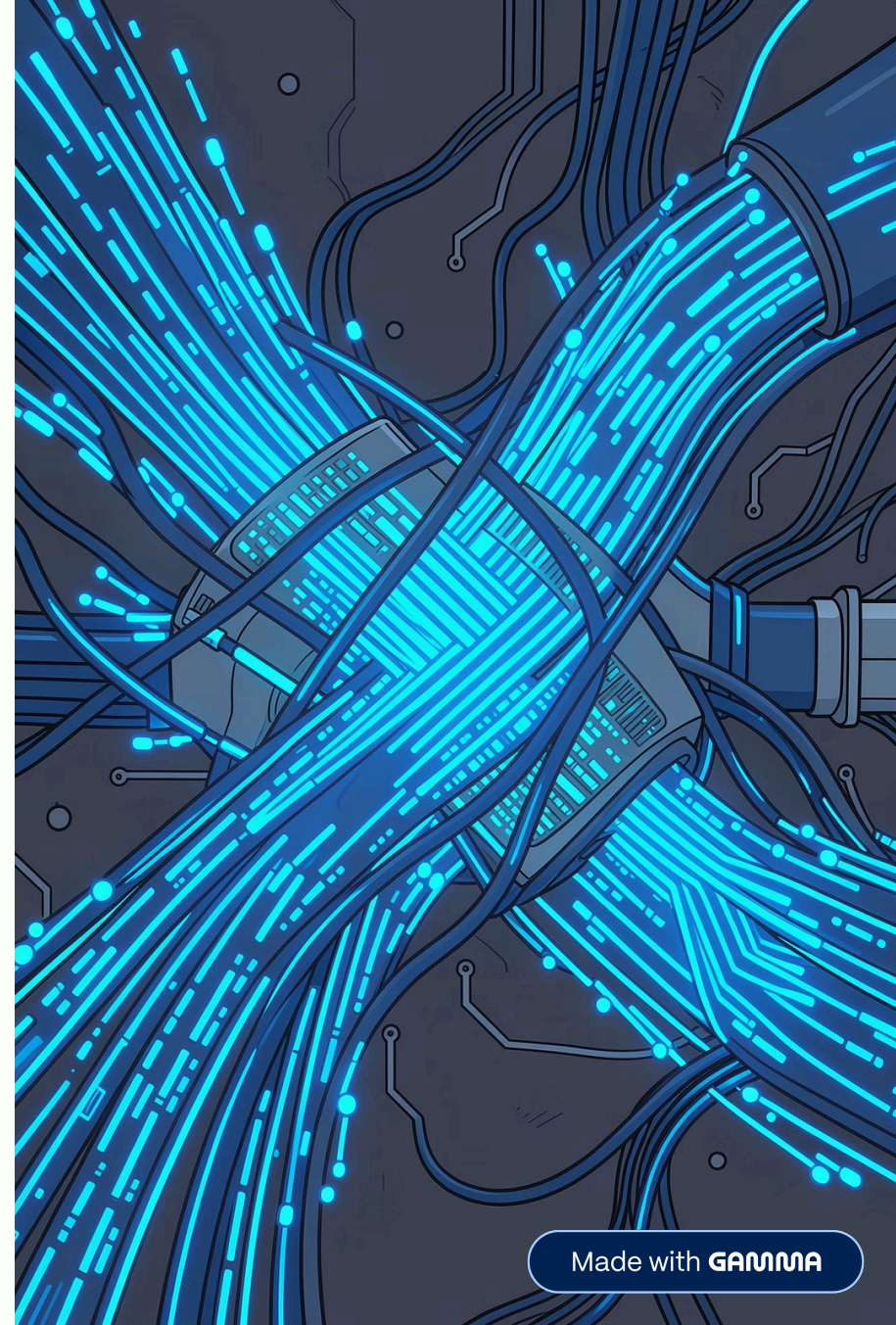


Telco Customer Churn Analysis

Business Analytics Case Study | Excel-Based Analysis

GIFT · ANALYSTLAB AFRICA INTERNSHIP · JUNE 2026



Made with GAMMA

The Scale of the Problem

The company is losing **more than 1 in 4 customers** – a structural challenge with clear, addressable drivers.

7,043

Total Customers

26.5%

Churn Rate

1,869 churned customers

\$139K

Monthly Revenue at Risk

5,174

Retained Customers

 Without intervention, the business risks continued revenue erosion, higher acquisition costs, and deteriorating customer lifetime value.

Dataset Overview

7,043 customer records across four data categories, analyzed entirely in Microsoft Excel.

AutoSave

Off

Telco-Customer-Chur... + Saved to this PC

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpTable Design

Paste

Clipboard

Font

Alignment

Number

Aptos Narrow11A^AA^A

General

FontColor

FontColor

Cell

customerID

Gender

SeniorCitizen

Partner

Dependents

PhoneService

MultipleLines

InternetService

OnlineSecurity

OnlineBackup

DeviceProtection

590-WHEO

Female

0

Non Senior Citizen

Yes

Partner

No

1

No phone service

DSL

No

Yes

No

375-WJEC

Male

0

Non Senior Citizen

No

No Partner

No

24

Yes

No

DSL

Yes

No

Yes

8568-QYBK

Male

0

Non Senior Citizen

No

No Partner

No

2

Yes

No

DSL

Yes

Yes

No

No

795-CFOOW

Male

0

Non Senior Citizen

No

No Partner

No

45

No

No phone service

DSL

Yes

No

No

Yes

837-HQIU

Female

0

Non Senior Citizen

No

No Partner

No

2

Yes

No

Fiber optic

No

No

No

No

8905-CDSKC

Female

0

Non Senior Citizen

No

No Partner

No

8

Yes

Yes

Fiber optic

No

No

No

Yes

452-KDWK

Male

0

Non Senior Citizen

No

No Partner

Yes

22

Yes

Yes

Fiber optic

No

Yes

No

No

1713-KOKNC

Female

0

Non Senior Citizen

No

No Partner

No

10

No

No phone service

DSL

Yes

No

No

No

1892-POOKP

Female

0

Non Senior Citizen

Yes

Partner

No

28

Yes

Yes

Fiber optic

No

No

Yes

No

3885-TABOU

Male

0

Non Senior Citizen

No

No Partner

Yes

62

Yes

No

DSL

Yes

Yes

No

No

0763-GRKD

Male

0

Non Senior Citizen

Yes

Partner

Yes

13

Yes

No

DSL

Yes

No

Yes

No

1469-LKBC

Male

0

Non Senior Citizen

No

No Partner

No

16

Yes

No

No

No internet service

No internet service

No internet service

No internet service

8961-TTAX

Male

0

Non Senior Citizen

Yes

Partner

No

58

Yes

Yes

Fiber optic

No

No

No

Yes

8280-LJEL

Male

0

Non Senior Citizen

No

No Partner

No

40

Yes

Yes

Fiber optic

No

Yes

Yes

Yes

1129-ALPS

Male

0

Non Senior Citizen

No

No Partner

No

25

Yes

No

Fiber optic

Yes

No

Yes

Yes

8555-SNQVZ

Female

0

Non Senior Citizen

Yes

Partner

Yes

69

Yes

Yes

Fiber optic

Yes

Yes

Yes

Yes

Yes

1391-KXVZB

Female

0

Non Senior Citizen

No

No Partner

No

52

No

No

No

No internet service

No internet service

No internet service

No internet service

859-WORKT

Male

0

Non Senior Citizen

No

No Partner

Yes

71

Yes

Yes

Fiber optic

Yes

No

No

Yes

1180-MFLUW

Female

0

Non Senior Citizen

Yes

Partner

Yes

10

Yes

 No | DSL | No | No | No | Yes |

1183-MYRS

Female

0

Non Senior Citizen

No

No Partner

No

21 Yes | No | Fiber optic | No | Yes | No | No |

1779-QEOMV

Male

1

Senior Citizen

No

No Partner

No

1 No | No phone service | DSL | No | No | No | Yes |

860-VOCOW

Male

0

Non Senior Citizen

Yes

Partner

No

12 Yes | No | No | No | No internet service | No internet service | No internet service |

856-WJEC

Male

0

Non Senior Citizen

No

No Partner

No

1 Yes | No | No | No | No internet service | No internet service | No internet service |

8538-WEABW

Female

0

Non Senior Citizen

Yes

Partner

No

58 Yes | Yes | DSL | No | Yes | No | No |

1222-MRPA

Male

0

Non Senior Citizen

Yes

Partner

Yes

48 Yes | No | DSL | Yes | Yes | Yes | No |

8655-QWJD

Female

0

Non Senior Citizen

No

No Partner

No

30 Yes | No | DSL | Yes | No | No | No |

1467-CHZW

Male

0

Non Senior Citizen

Yes

Partner

Yes

47 Yes | Yes | Fiber optic | No | Yes | No | No |

865-UTDMZ

Male

0

Non Senior Citizen

Yes

Partner

Yes

1 No | No phone service | DSL | No | Yes | No | No |

848-VJUN

Male

0

Non Senior Citizen

Yes

Partner

No

72 Yes | Yes | DSL | Yes | Yes | Yes | No |

1773-HWJUZ

Female

0

Non Senior Citizen

No

No Partner

Yes

17 Yes | No | DSL | No | No | No | No |

841-WFECA

Female

1

Senior Citizen

Yes

Partner

No

71 Yes | Yes | Fiber optic | Yes | Yes | Yes | Yes |

869-WHWW

Male

1

Senior Citizen

Yes

Partner

No

2 Yes | No | Fiber optic | No | No | No | No |

127-EAUQ

Female

0

Non Senior Citizen

Yes

Partner

Yes

27 Yes | No | DSL | Yes | Yes | Yes | Yes |

Demographics

Gender, Senior Citizen, Partner, Dependents

Account Details

Tenure, Contract Type, Payment Method

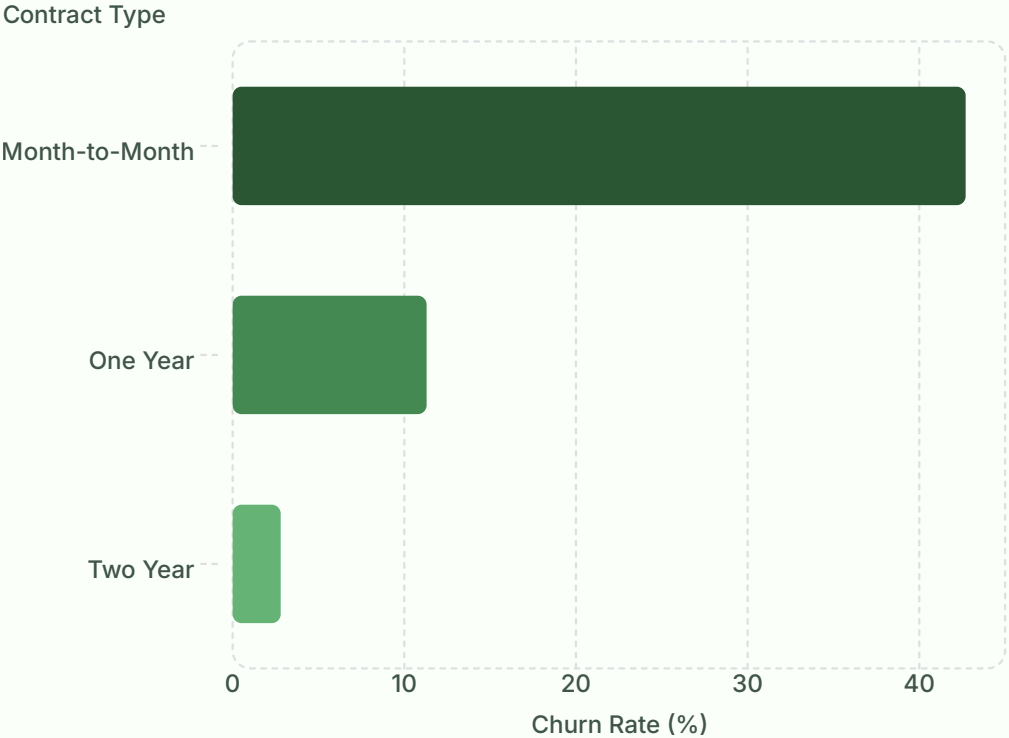
Services Used

Phone, Internet, Streaming, Security, Tech Support

Financials

Monthly Charges, Total Charges

Top Finding: Contract Type Is the Strongest Predictor



15× Higher Churn

Month-to-month customers churn at **42.7%** – 15 times the rate of two-year contract holders. This pattern holds across every dimension analyzed.

~1,655

Month-to-month churned

~166

One-year churned

~48

Two-year churned

Tenure & Internet Service: Critical Risk Zones

Tenure Churn Trend



Key Insights

Early months are critical: Churn exceeds 47% in the first year, falling below 7% for customers with 60 - 72 months tenure. Churned customers average only 18 months vs. 37.7 for retained.

Fiber optic paradox: Despite being the premium product, fiber optic customers churn at **41.9%** – more than twice the DSL rate (19.0%).

Demographics & Payment Behavior

Senior Citizens

41.7% churn rate vs. 23.7% for non-seniors – potentially reflecting affordability or service complexity concerns.

Customers Without Partner

33.0% churn rate vs. 19.7% for those with a partner. Without dependents: 31.3% vs. 15.5%.

Electronic Check Payers

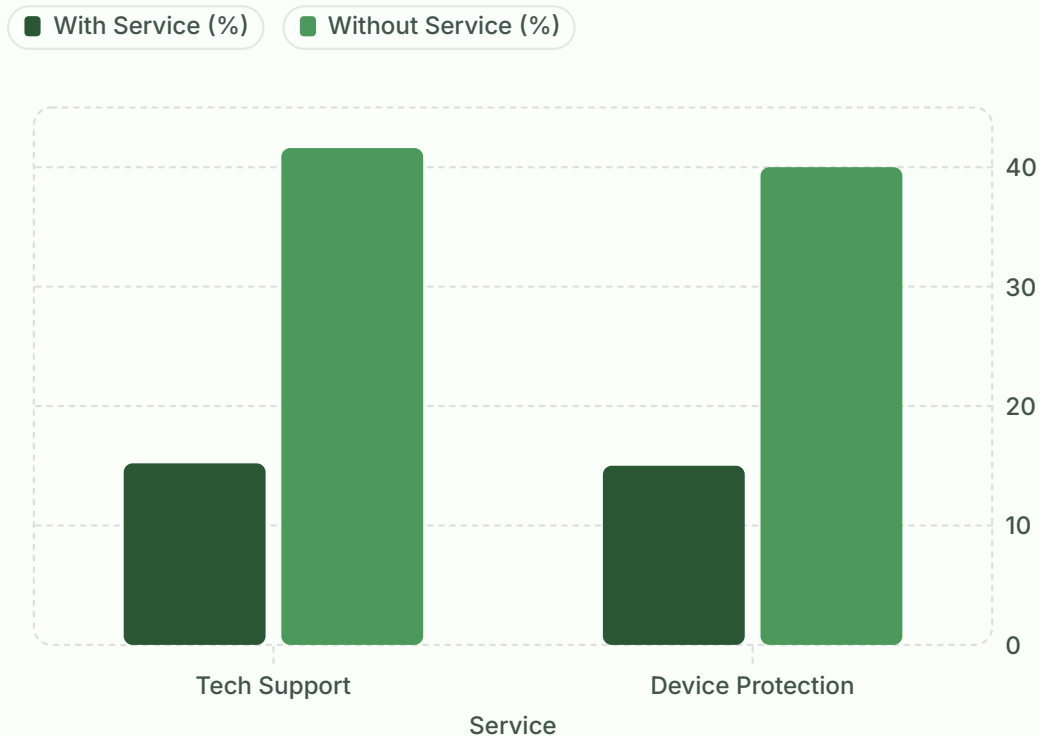
45.3% churn rate – the highest of any payment method. 1,071 churned customers paid this way vs. ~15-16% for auto-pay users.

Paperless Billing

33.6% churn rate – double the 16.4% rate of paper billing customers, likely overlapping with the high-risk month-to-month segment.



Add-On Services: The Loyalty Anchor



Add-On Services Reduce Churn by ~65%

Customers subscribed to Tech Support churn at just **15.2%** vs. 41.6% without it. Device Protection shows a similar pattern.

These services act as **loyalty anchors** – bundling them more aggressively could significantly lift overall retention.

- ✓ Cross-selling add-ons during onboarding and renewal is one of the highest-ROI interventions available.

Business Insights Summary

Patterns Discovered

- Churn is **structural, not random** – contract type, tenure, and service type create predictable risk tiers.
- Risk **declines linearly with time** – every additional year roughly halves churn probability.
- Auto-pay users churn at **one-third the rate** of manual payers – a reliable commitment proxy.

Highest-Risk Segments

Month-to-Month

42.7% churn – largest volume

New Customers (0–12 mo)

47.7% churn – onboarding gap

Fiber Optic

41.9% churn – premium product

Electronic Check

45.3% churn – highest payment risk

Five Strategic Recommendations

01

Contract Migration Incentive

Reward month-to-month customers who upgrade to annual/two-year contracts. **Expected impact: HIGH**

02

First-Year Onboarding Journey

90-day program with touchpoints at Day 7, 30, 60. Drive add-on adoption early. **Expected impact: HIGH**

03

Auto-Pay Conversion Campaign

Offer \$3 - \$5/month credit to switch electronic check users to auto-pay. **Expected impact: MEDIUM-HIGH**

04

Address Fiber Optic Dissatisfaction

Survey fiber customers, fix pricing or reliability issues, consider loyalty rate reductions. **Expected impact: HIGH**

05

Bundle Add-On Services

Discounted bundles with Tech Support or Device Protection; 30-day free trials for high-risk customers. **Expected impact: MEDIUM**

Conclusion & Key Takeaways

The Path Forward

Churn is not random – it follows predictable, addressable patterns. Implementing these five recommendations in sequence offers a realistic path to reducing churn by **8-12 percentage points**, recovering hundreds of thousands in monthly recurring revenue.

Key Takeaways at a Glance

7,043 customers analyzed · **1,869** churned (26.5%)

\$139,131 monthly revenue at risk

Month-to-month (**42.7%**) and first-year customers (**47.7%**) are highest-risk

Add-on services reduce churn by up to **65%**

Auto-pay adoption is the **simplest, fastest win** available